

Week 3 Assignment

1. Imported superstore dataset, created 3 tables and inserted data into them using SELECT DISTINCT

DESCRIBE superstore_raw;

	Field	Type	Null	Key	Default	Extra
►	Row ID	int	YES		NULL	
	Order ID	text	YES		NULL	
	Order Date	text	YES		NULL	
	Ship Date	text	YES		NULL	
	Ship Mode	text	YES		NULL	
	Customer ID	text	YES		NULL	
	Customer Name	text	YES		NULL	
	Segment	text	YES		NULL	
	Country	text	YES		NULL	
	City	text	YES		NULL	
	State	text	YES		NULL	
	Postal Code	int	YES		NULL	
	Region	text	YES		NULL	
	Product ID	text	YES		NULL	
	Category	text	YES		NULL	
	Sub-Category	text	YES		NULL	
	Product Name	text	YES		NULL	
	Sales	double	YES		NULL	
	Quantity	int	YES		NULL	
	Discount	double	YES		NULL	
	Profit	double	YES		NULL	

```
CREATE TABLE customers AS
SELECT DISTINCT `Customer ID`, `Customer Name`
FROM superstore_raw;
```

```
CREATE TABLE orders AS
SELECT DISTINCT `Order ID`, `Customer ID`, `Order Date`, `Sales`
FROM superstore_raw;
```

```
CREATE TABLE products AS
SELECT DISTINCT `Product ID`, `Product Name`, `Category`
FROM superstore_raw;
```

✓	12	23:14:12	DESCRIBE superstore_raw	21 row(s) returned	0.000 sec / 0.000 sec
✓	13	23:15:13	CREATE TABLE customers AS SELECT DISTINCT `Customer ID`, `Customer Nam...	793 row(s) affected Records: 793 Duplicates: 0 Warnings: 0	0.062 sec
✓	14	23:15:21	CREATE TABLE orders AS SELECT DISTINCT `Order ID`, `Customer ID`, `Order D...	9692 row(s) affected Records: 9692 Duplicates: 0 Warnings: 0	0.421 sec
✓	15	23:15:23	CREATE TABLE products AS SELECT DISTINCT `Product ID`, `Product Name`, `...	1842 row(s) affected Records: 1842 Duplicates: 0 Warnings: 0	0.094 sec

Steps 2 and 3

1. **Orders Greater Than Average Sales (Subquery):**

```
SELECT * FROM orders
WHERE `Sales` > (SELECT AVG(`Sales`) FROM orders);
```

	Order ID	Customer ID	Order Date	Sales
▶	CA-2016-152156	CG-12520	11/8/2016	261.96
	CA-2016-152156	CG-12520	11/8/2016	731.94
	US-2015-108966	SO-20335	10/11/2015	957.5775
	CA-2014-115812	BH-11710	6/9/2014	907.152
	CA-2014-115812	BH-11710	6/9/2014	1706.184
	CA-2014-115812	BH-11710	6/9/2014	911.424
	CA-2016-161389	IM-15070	12/5/2016	407.976
	CA-2014-105893	PK-19075	11/11/2014	665.88
	CA-2015-106320	EB-13870	9/25/2015	1044.63
	US-2015-150630	TB-21520	9/17/2015	3083.43
	CA-2016-117590	GH-14485	12/8/2016	1097.544
	CA-2015-117415	SN-20710	12/27/2015	532.3992
	CA-2015-117415	SN-20710	12/27/2015	371.168
	CA-2016-105816	JM-15265	12/11/2016	1029.95
	CA-2016-111682	TB-21055	6/17/2016	319.41
	CA-2014-106376	BS-11590	12/5/2014	1113.024
	US-2015-134026	JE-15745	4/26/2015	831.936
	CA-2017-140088	PO-18865	5/28/2017	301.96
	US-2015-156867	LC-16870	11/13/2015	238.896
	CA-2015-129476	PA-19060	10/15/2015	339.96
	CA-2015-110457	DK-13090	3/2/2015	787.53
	US-2014-152030	AD-10180	12/26/2014	600.558
	US-2014-134614	PF-19165	9/20/2014	617.7
	US-2017-107272	TS-21610	11/5/2017	243.887

2. Highest Sales Order per Customer (Subquery):

```
SELECT `Customer ID`, MAX(`Sales`) AS highest_order
FROM orders
GROUP BY `Customer ID`;
```

	Customer ID	highest_order
▶	CG-12520	731.94
	DV-13045	721.875
	SO-20335	957.5775
	BH-11710	1706.184
	AA-10480	479.97
	IM-15070	1819.86
	HP-14815	1242.9
	PK-19075	4305.552
	AG-10270	1336.44
	ZD-21925	823.96
	KB-16585	1013.832
	SF-20065	254.9
	EB-13870	1044.63
	EH-13945	100.704
	TB-21520	3083.43
	MA-17560	799.984
	GH-14485	1097.544
	SN-20710	532.3992
	LC-16930	2396.4
	RA-19885	1215.92
	ES-14080	1319.8
	ON-18715	630.024
	PO-18865	604.752
	IM-16800	638.87

3. Total Sales for Each Customer (CTE):

```
WITH CustTotal AS (  
    SELECT `Customer ID`, SUM(`Sales`) AS total_s  
    FROM orders GROUP BY `Customer ID`  
)  
SELECT * FROM CustTotal;
```

	Customer ID	total_s
▶	CG-12520	1148.7800000000002
	DV-13045	1119.4830000000002
	SO-20335	2602.5755000000001
	BH-11710	6255.3510000000001
	AA-10480	1782.872
	IM-15070	4930.4739999999999
	HP-14815	1990.314
	PK-19075	8158.654
	AG-10270	2565.2579999999994
	ZD-21925	1493.944
	KB-16585	2742.7580000000003
	SF-20065	763.5459999999998
	EB-13870	2767.219
	EH-13945	684.171
	TB-21520	4730.628
	MA-17560	4280.6209999999999
	GH-14485	1361.2440000000001
	SN-20710	3127.9592000000007
	LC-16930	4481.162
	RA-19885	3832.3139999999994
	ES-14080	4648.516
	ON-18715	1567.52
	PO-18865	2493.2140000000004
	HA-16900	7285.3320000000003

4. Customers with Total Sales Above Average (CTE + Subquery):

```
WITH CustTotal AS (  
    SELECT `Customer ID`, SUM(`Sales`) AS ts FROM orders GROUP BY `Customer ID`  
)  
SELECT * FROM CustTotal  
WHERE ts > (SELECT AVG(ts) FROM CustTotal);
```

	Customer ID	ts
▶	BH-11710	6255.3510000000001
	IM-15070	4930.4739999999999
	PK-19075	8158.654
	TB-21520	4730.628
	MA-17560	4280.6209999999999
	SN-20710	3127.9592000000007
	LC-16930	4481.162
	RA-19885	3832.3139999999994
	ES-14080	4648.516
	KM-16720	4909.4720000000001
	KD-16270	8282.358
	JE-15745	6400.4549999999999
	SC-20770	4492.6569999999999
	CS-12400	3651.8585000000007
	GM-14455	3078.6240000000003
	RB-19465	4556.5660000000002
	GZ-14470	4355.15
	JM-15250	3145.3740000000003
	CL-12565	10880.545999999998
	SC-20725	5226.2099999999998
	AD-10180	6100.9999999999999
	LS-16975	2988.674
	LC-16885	7396.7739999999985
	DA-15805	6456.456

5. Rank Customers Based on Total Sales (Window Function):

```
SELECT `Customer ID`, SUM(`Sales`) AS total_sales,
RANK() OVER (ORDER BY SUM(`Sales`) DESC) AS sales_rank
FROM orders
GROUP BY `Customer ID`;
```

	Customer ID	total_sales	sales_rank
▶	SM-20320	25043.05	1
	TC-20980	19017.847999999998	2
	RB-19360	15117.339	3
	TA-21385	14595.62	4
	AB-10105	14355.610999999997	5
	SC-20095	14142.333999999999	6
	KL-16645	14071.917	7
	HL-15040	12873.297999999999	8
	SE-20110	12209.438000000002	9
	CC-12370	12129.072	10
	TS-21370	11885.871	11
	GT-14710	11820.119999999997	12
	BM-11140	11609.900000000001	13
	SV-20365	11420.645999999997	14
	CJ-12010	11079.742	15
	CL-12565	10880.545999999998	16
	ME-17320	10635.917999999998	17
	KF-16285	10604.266	18
	BS-11365	10501.652999999998	19
	EH-13765	10005.980000000001	20
	JL-15835	9403.003	21
	GT-14635	9351.211999999998	22
	HW-14935	9300.253999999997	23
	TS-21400	8812.744000000000	24

6. Row Numbers per Order within Customer (Window Function + PARTITION BY):

```
SELECT `Customer ID`, `Order ID`, `Order Date`,
ROW_NUMBER() OVER (PARTITION BY `Customer ID` ORDER BY `Order Date`) AS row_num
FROM orders;
```

	Customer ID	Order ID	Order Date	row_num
▶	AA-10315	CA-2015-121391	10/4/2015	1
	AA-10315	CA-2016-103982	3/3/2016	2
	AA-10315	CA-2016-103982	3/3/2016	3
	AA-10315	CA-2016-103982	3/3/2016	4
	AA-10315	CA-2016-103982	3/3/2016	5
	AA-10315	CA-2014-128055	3/31/2014	6
	AA-10315	CA-2014-128055	3/31/2014	7
	AA-10315	CA-2017-147039	6/29/2017	8
	AA-10315	CA-2017-147039	6/29/2017	9
	AA-10315	CA-2014-138100	9/15/2014	10
	AA-10315	CA-2014-138100	9/15/2014	11
	AA-10375	CA-2014-130729	10/24/2014	1
	AA-10375	CA-2015-114503	11/13/2015	2
	AA-10375	CA-2016-131065	11/14/2016	3
	AA-10375	CA-2016-131065	11/14/2016	4
	AA-10375	CA-2016-131065	11/14/2016	5
	AA-10375	CA-2017-100230	12/11/2017	6
	AA-10375	CA-2017-100230	12/11/2017	7
	AA-10375	CA-2017-100230	12/11/2017	8
	AA-10375	CA-2015-140921	2/3/2015	9
	AA-10375	CA-2015-140921	2/3/2015	10
	AA-10375	CA-2014-158064	4/21/2014	11
	AA-10375	CA-2015-109939	5/8/2015	12
	AA-10375	CA-2016-126613	7/10/2016	13

7. Top 3 Customers Based on Total Sales (Window Function):

```
SELECT `Customer ID`, SUM(`Sales`) AS total_s
FROM orders
GROUP BY `Customer ID`
ORDER BY total_s DESC LIMIT 3;
```

	Customer ID	total_s
▶	SM-20320	25043.05
	TC-20980	19017.847999999998
	RB-19360	15117.339

8. Final Combined Query:

```
WITH CustomerTotals AS (
  SELECT `Customer ID`, SUM(`Sales`) AS total_sales
  FROM orders
  GROUP BY `Customer ID`
)
SELECT
  c.`Customer Name`,
  ct.total_sales,
  RANK() OVER (ORDER BY ct.total_sales DESC) AS sales_rank
FROM customers c
JOIN CustomerTotals ct ON c.`Customer ID` = ct.`Customer ID`;
```

	Customer Name	total_sales	sales_rank
▶	Sean Miller	25043.05	1
	Tamara Chand	19017.847999999998	2
	Raymond Buch	15117.339	3
	Tom Ashbrook	14595.62	4
	Adrian Barton	14355.610999999997	5
	Sanjit Chand	14142.333999999999	6
	Ken Lonsdale	14071.917	7
	Hunter Lopez	12873.297999999999	8
	Sanjit Engle	12209.438000000002	9
	Christopher Conant	12129.072	10
	Todd Sumrall	11885.871	11
	Greg Tran	11820.119999999997	12
	Becky Martin	11609.900000000001	13
	Seth Vernon	11420.645999999997	14
	Caroline Jumper	11079.742	14
	Clay Ludtke	10880.545999999998	16
	Maria Etezadi	10635.917999999998	17
	Karen Ferguson	10604.266	18
	Bill Shonely	10501.652999999998	19
	Edward Hooks	10005.980000000001	20
	John Lee	9403.003	21
	Grant Thornton	9351.211999999998	22
	Helen Wasserman	9300.253999999997	23
	Tom Boeckenhauer	9012.741999999998	24

Mini Project

1. Top 5 Customers:

```
SELECT `Customer Name`, SUM(`Sales`) AS total_sales
FROM customers c
JOIN orders o ON c.`Customer ID` = o.`Customer ID`
GROUP BY `Customer Name`
ORDER BY total_sales DESC LIMIT 5;
```

	Customer Name	total_sales
▶	Sean Miller	25043.05
	Tamara Chand	19017.847999999998
	Raymond Buch	15117.339
	Tom Ashbrook	14595.62
	Adrian Barton	14355.610999999997

2. Bottom 5 Customers:

```
SELECT `Customer Name`, SUM(`Sales`) AS total_sales
FROM customers c
JOIN orders o ON c.`Customer ID` = o.`Customer ID`
GROUP BY `Customer Name`
ORDER BY total_sales ASC LIMIT 5;
```

	Customer Name	total_sales
▶	Thais Sissman	4.833
	Lela Donovan	5.304
	Mitch Gastineau	12.32
	Carl Jackson	16.52
	Roy Skaria	22.328

3. Customers with only one order:

```
SELECT `Customer ID`, COUNT(`Order ID`) AS order_count
FROM orders
GROUP BY `Customer ID`
HAVING order_count = 1;
```

	Customer ID	order_count
▶	JR-15700	1
	MG-18205	1
	AO-10810	1
	LD-16855	1
	CJ-11875	1
	RE-19405	1

4. Customers with above-average sales:

```
WITH CustomerTotals AS (  
    SELECT `Customer ID`, SUM(`Sales`) AS total_sales FROM orders GROUP BY `Customer ID`  
)  
SELECT * FROM CustomerTotals  
WHERE total_sales > (SELECT AVG(total_sales) FROM CustomerTotals);
```

	Customer ID	total_sales
▶	BH-11710	6255.351000000001
	IM-15070	4930.473999999999
	PK-19075	8158.654
	TB-21520	4730.628
	MA-17560	4280.620999999999
	SN-20710	3127.9592000000007
	LC-16930	4481.162
	RA-19885	3832.3139999999994
	ES-14080	4648.516
	KM-16720	4909.472000000001
	KD-16270	8282.358
	JE-15745	6400.454999999999
	SC-20770	4492.656999999999
	CS-12400	3651.8585000000007
	GM-14455	3078.6240000000003
	RB-19465	4556.5660000000002
	GZ-14470	4355.15
	JM-15250	3145.3740000000003
	CL-12565	10880.545999999998
	SC-20725	5226.2099999999998
	AD-10180	6100.999999999999
	LS-16975	2988.674

5. Highest order value per customer:

```
SELECT `Customer ID`, MAX(`Sales`) AS highest_order  
FROM orders  
GROUP BY `Customer ID`;
```

	Customer ID	highest_order
▶	CG-12520	731.94
	DV-13045	721.875
	SO-20335	957.5775
	BH-11710	1706.184
	AA-10480	479.97
	IM-15070	1819.86
	HP-14815	1242.9
	PK-19075	4305.552
	AG-10270	1336.44
	ZD-21925	823.96
	KB-16585	1013.832
	SF-20065	254.9
	EB-13870	1044.63
	EH-13945	100.704
	TB-21520	3083.43
	MA-17560	799.984
	GH-14485	1097.544
	SN-20710	532.3992
	LC-16930	2396.4
	RA-19885	1215.92
	ES-14080	1319.8
	ON-18715	630.024
	PO-18865	604.752